

For full credit explain your reasoning, showing all relevant work.

Exercise 1. Recall that if G is a group and $a \in G$, then $|a|$ denotes the order of a .

a) Show that for any $a, b \in G$ we have $|a| = |bab^{-1}|$.

b) Show that for any $a, b \in G$ we have $|ab| = |ba|$.

Exercise 2. Let G be a finite group such that $|G|$ is even. Show that the number of elements of order 2 in G is odd. In particular, since 0 is an even number, G contains at least one element of order 2.

Exercise 3. Let D_3 be the dihedral group of symmetries of an equilateral triangle.

a) Let $A \subseteq D_3$ be a subset consisting of third powers of elements of D_3 :

$$A = \{g^3 \in D_3 \mid g \in D_3\}$$

Is A a subgroup of D_3 ? Justify your answer.

b) Let $B \subseteq D_3$ be a subset consisting of squares of elements of D_3 :

$$B = \{g^2 \in D_3 \mid g \in D_3\}$$

Is B a subgroup of D_3 ? Justify your answer.

Hint: See section 5 of the lecture notes for a description of elements of groups D_n .

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Let G be a group and let $g \in G$. Show that $|g| = |g^{-1}|$.

Practice Exercise 2. Let G be a finite group and let $n > 2$. Show that the number of elements of G of order n is even.

Practice Exercise 3. Assume that in a group G there exists an element a of order pk where p is a prime number. Let $b \in G$ be an element such that $b^p = a$. Show that $|b| = p^2k$.

Practice Exercise 4. Let G be a group and let A_1, A_2 be two subgroups of G . Define a subset $A_1A_2 \subseteq G$ by

$$A_1A_2 = \{a_1a_2 \mid a_1 \in A_1, a_2 \in A_2\}$$

a) Consider the dihedral group D_4 :

$\circ$	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
I	I	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	I	D'	D	H	V
R_{180}	R_{180}	R_{270}	I	R_{90}	V	H	D'	D
R_{270}	R_{270}	I	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	I	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	I	R_{270}	R_{90}
D	D	H	D'	V	R_{270}	R_{90}	I	R_{180}
D'	D'	V	D	H	R_{90}	R_{270}	R_{180}	I

Take the subgroups of D_4 given by $A_1 = \{I, D\}$ and $A_2 = \{I, H\}$. Which elements are in the set A_1A_2 ? Which elements are in the set A_2A_1 ? Is the set A_1A_2 a subgroup of D_4 ?

b) Show that if A_1, A_2 are subgroups of a group G such that $A_1A_2 = A_2A_1$ then A_1A_2 is a subgroup of G .